

Interpretability and Text-Diff Methodology

KJV 1611 / KJV 1769 Semantic Drift Study

Purpose

This document defines the interpretability methodology used to translate high drift text unit pairs into human-readable explanations. It supports the qualitative validation of thresholded outliers and prepares category data for enrichment analysis.

Core Principle

Embedding distance is not self-explanatory. Every flagged unit must be supported by structured textual comparison, categorized explanation, and reproducible transformation logic. This preserves interpretability and prevents overstatement of model generated distance values.

Input Sets

The primary interpretability input consists of STRONG_OUTLIER units. REVIEW_CANDIDATE units are processed using the same methodology as a broader exploratory set. Both sets retain raw and normalized text views, drift values, residual fields, and threshold classifications.

Required Text-Diff Layers

Layer	Purpose	Examples
Character-level diff	Captures fine grained orthographic, punctuation, capitalization, and spacing variation.	vnto → unto; Iesus → Jesus; : → ;
Token-level diff	Captures word insertions, deletions, substitutions, and preservation.	euening hee commeth → evening he cometh
Phrase-level diff	Captures multi-token and structural changes.	Clause-level resegmentation or reordered phrases

HITL Category Schema

The annotation schema is multi-label.

Category	Operational Definition	Representative Example / Cue
orthographic_character_form	Spelling, historical character form, capitalization, abbreviation, ampersand expansion, or form level modernization that preserves lexical meaning and syntactic function.	Iesus → Jesus; vnto → unto; & → and
punctuation_clause_relation	A punctuation change that may alter clause relation, scope, coordination, explanation, contrast, continuation, or syntactic attachment without necessarily changing the lexical words.	: → ,; ; → ,, , → ., or other punctuation changes that affect whether a clause reads as explanation, continuation, coordination, apposition, or a new syntactic unit.
lexical_substitution	One word or phrase is replaced with a different lexical item that could affect	A changed noun, verb, adjective, title, theological term, or phrase where the

	nuance, specificity, reference, or meaning.	replacement is not merely spelling modernization.
function_word_change	Changes to articles, prepositions, conjunctions, particles, or other grammatical function markers.	in → into, of → from, and → but, added/removed the, a, to, for, or similar grammatical-word changes.
pronoun_shift	A change involving pronouns, possessive pronouns, antecedent clarity, person, number, gender reference, or pronominal ambiguity.	he → they, his → their, thou/thee/ye/you changes, or a pronoun becoming more or less ambiguous.
negation_modality_change	A change involving negation, obligation, permission, possibility, certainty, command, tense/aspect force, or modal meaning.	shall, will, may, might, must, cannot, not, never, no, or changes affecting imperative, conditional, prohibitive, or certainty force.
word_order_syntactic_shift	Alterations in the sequence or syntactic ordering of constituents.	The same basic words are present, but their order, grouping, or syntactic attachment changes.
addition_deletion	Material is added, removed, or substantially expanded/contracted compared with the paired edition.	A word, phrase, clause, name, title, explanatory element, or repeated phrase appears on one side but not the other.
multi_factor_complex	Overlapping or compounded changes where multiple categories interact within a single unit.	Mixed lexical, syntactic, and punctuation changes

Residual Drift Classification

Residual behavior is interpreted by comparing R0 and R1 drift. R0-high/R1-low patterns suggest surface form resolution by normalization. R0-high/R1-high patterns indicate persistent structural or semantic distance. Unexpected R1 increases are flagged for artifact or model sensitivity review.

Category Enrichment Preparation

The interpretability dataset supports contingency table analysis of category presence in the flagged tail relative to the inlier background sample. Fisher exact testing or chi-square testing is applied depending on cell density, and p-values are BH-FDR adjusted for multiple comparisons.

Interpretation Safeguard

Interpretability analysis identifies types of textual difference. It does not establish doctrinal conclusions without additional philological and theological review.

Study Decisions

Component	Study Decision
Primary input	STRONG OUTLIER units
Secondary input	REVIEW CANDIDATE units
Diff layers	Character, token, and phrase
Category system	Multi label HITL taxonomy
H3 support	Category enrichment dataset
Interpretive boundary	Textual difference categories. No standalone doctrinal conclusions

Expected Output Artifacts

- interpretability_dataset.csv
- top_outliers_annotated.csv
- category_annotation_dataset.csv
- category_counts.csv
- category_enrichment_results.csv